Experiment: 7

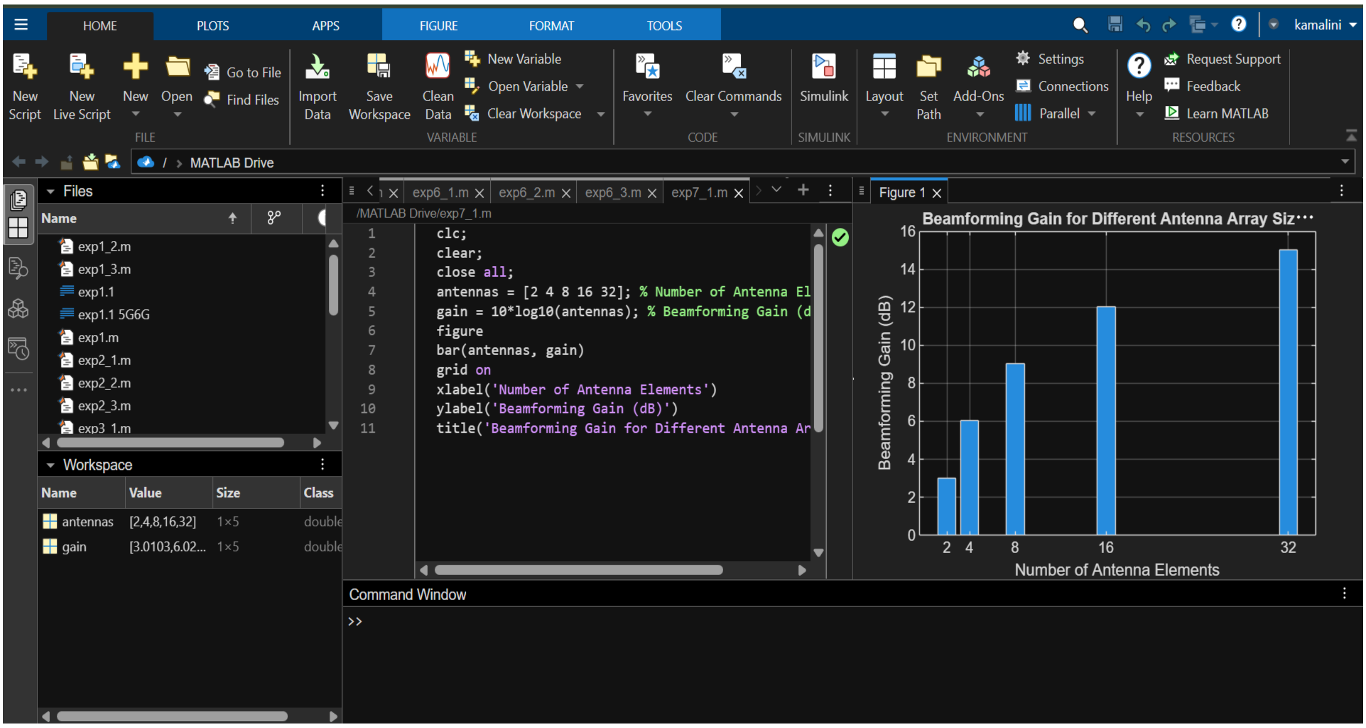
Analog Beamforming Analysis Using MATLAB

Objectives 1:

To analyze the beamforming gain achieved by varying the number of antenna elements in an analog beamforming system using MATLAB.

CODE:

```
clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

gain = 10*log10(antennas); % Beamforming Gain (dB)

figure

bar(antennas, gain)

grid on

xlabel('Number of Antenna Elements')

ylabel('Beamforming Gain (dB)')

title('Beamforming Gain for Different Antenna Array Sizes')
```

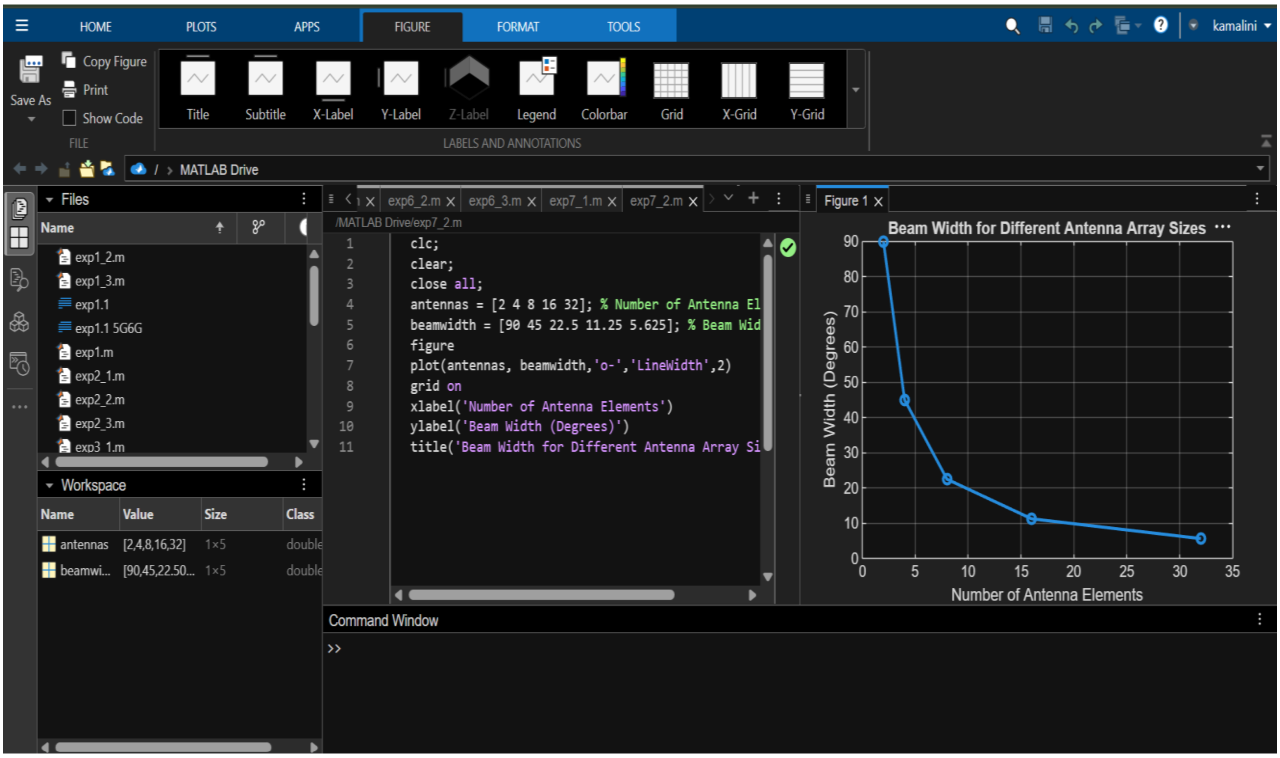


Objectives 2:

To compare the beam width of an antenna array for different beamforming configurations using MATLAB.

CODE:

```
clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)

figure

plot(antennas, beamwidth,'o-','LineWidth',2)

grid on

xlabel('Number of Antenna Elements')

ylabel('Beam Width (Degrees)')

title('Beam Width for Different Antenna Array Sizes')
```



Objectives 3:

To analyze the main lobe direction by steering the antenna array beam to different angles using MATLAB.

CODE:

```
clc;

clear;

close all;

theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)

gain = [1 1 1 1 1]; % Constant normalized gain

figure

polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)

title('Main Lobe Direction in Analog Beamforming')
```